WEEK 4

Superposition, Entanglement, and Bell's Theorem

8, 10 September · Objective 2

Superposition is the resource; entanglement is the correlation that has no classical explanation. Bell's theorem converts a philosophical dispute into an experimental one, and the experiment has been run. This is the week where "quantum is weird" becomes a measured number.

Superposition, precisely

A qubit state is a normalized complex combination of basis states:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle \qquad |\alpha|^2 + |\beta|^2 = 1$$

Three clarifications that prevent most downstream errors:

It is not "both 0 and 1." It is a definite vector in a two-dimensional complex space that happens not to be an eigenvector of the measurement operator you chose. In the basis $\{|+\rangle, |-\rangle\}$ where $|\pm\rangle = (|0\rangle \pm |1\rangle)/\sqrt{2}$, the state $|+\rangle$ is *not* superposed — it is a basis state with a certain outcome. **Superposition is basis-relative.** Entanglement is not.

It is not ignorance. A coin under a cup is 50/50 because you lack information; the coin has a value. Measuring $|+\rangle$ in the $\{|0\rangle, |1\rangle\}$ basis gives 50/50, but measuring in $\{|+\rangle, |-\rangle\}$ gives $|+\rangle$ with certainty. No probability distribution over hidden values reproduces both.

Global phase is unobservable; relative phase is everything. $e^{i\gamma}|\psi\rangle$ is physically identical to $|\psi\rangle$. But $(|0\rangle + |1\rangle)/\sqrt{2}$ and $(|0\rangle - |1\rangle)/\sqrt{2}$ are perfectly distinguishable — apply H to each and measure.

Analogy boundary — superposition as pending dispatch. An agent holding several candidate execution paths before commit is a good first model: multiple outcomes live, weights attached, one selected at dispatch. Two failures. First, your candidate set is basis-independent — the paths are the paths regardless of how you enumerate them; quantum "paths" are an artifact of basis choice. Second, and fatally, your weights are probabilities that only ever add. Amplitudes cancel. An agent whose two live paths can annihilate on merge has no software analogue, and that cancellation is where every quantum speedup comes from.

Entanglement

For two qubits, the state space is the tensor product — four dimensions, spanned by |00⟩, |01⟩, |10⟩, |11⟩. A state is **separable** if it factors:

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|ψ⟩ = |a⟩ ⊗ |b⟩
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and **entangled** otherwise. The canonical example is a Bell state:

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|Φ⁺⟩ = (|00⟩ + |11⟩)/√2
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No |a⟩, |b⟩ produce this. Try it: expanding $(a_0|0\rangle + a_1|1\rangle)(b_0|0\rangle + b_1|1\rangle)$ requires $a_0b_1 = 0$ and $a_1b_0 = 0$ while a_0b_0 and a_1b_1 are both nonzero — inconsistent.

The consequence: each qubit individually is in a maximally mixed state, carrying no information. All the information lives in the correlation. Measure the first in any basis and the second is instantly determined in that same basis, at any separation.

Analogy boundary — entanglement as shared state. Reading a Bell pair as two references to one object captures the correlation and gets the mechanism backwards in a way that matters. Shared mutable state implies a channel — writes propagate, and you could signal. Entanglement **cannot** transmit information: the marginal distribution at each end is uniform regardless of what the other party does, which is the no-signaling theorem. The correlation is only visible when the two records are brought together and compared — which requires a classical channel at light speed. Closer to your domain: two nodes deterministically deriving the same value from a shared seed. That analogy is *too* good, and Bell's theorem is precisely the proof that no shared seed exists.

EPR and the completeness argument

Einstein, Podolsky, and Rosen, 1935. Take $|\Phi^+\rangle$ with the halves separated. Measure qubit A in the Z basis and you know B's Z value with certainty, without touching B. Measure A in X instead and you know B's X value with certainty.

Their reasoning: if you can predict a quantity with certainty without disturbing the system, that quantity corresponds to an "element of reality." Since you could have chosen either basis, and the choice at A cannot affect B (locality), B must possess definite values for *both* Z and X. But quantum mechanics assigns no such state. Therefore the description is **incomplete** — there are hidden variables.

This is a rigorous argument. Its premises are locality and realism, and its conclusion follows. Bohr's reply was, by wide agreement, obscure. For thirty years the dispute was considered untestable metaphysics.

Bell's theorem

Bell's 1964 insight: the hidden-variable hypothesis has **testable consequences**, and they differ from quantum mechanics.

Assume each particle carries variables λ fixing outcomes for every measurement setting, and that A's setting cannot influence B's outcome. Measure spin along directions a, a′ (at A) and b, b′ (at B), outcomes ±1. Define the correlation $E(a, b) = \langle A{\cdot}B\rangle$. Then the CHSH combination obeys:

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S = |E(a,b) - E(a,b′) + E(a′,b) + E(a′,b′)|  ≤  2
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The derivation needs nothing about quantum mechanics — only that a joint distribution over all four outcomes exists. It is essentially a counting argument.

Quantum mechanics predicts, for $|\Phi^+\rangle$ at optimal angles (45° apart):

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S = 2√2 ≈ 2.828       (Tsirelson's bound)
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Experiment agrees with quantum mechanics. Aspect closed the timing loophole in 1982; the 2015 loophole-free tests at Delft, NIST, and Vienna closed locality and detection together. Nobel Prize, 2022.

What this rules out: any theory that is both local and realist. Something has to go. Most physicists drop realism (no pre-existing values); Bohmians drop locality. Nothing forces the choice, but the conjunction is dead.

> **Analogy boundary — the shared-seed model.** Two nodes with a common seed produce identical outputs with no communication and perfectly mimic entanglement for *one* fixed measurement setting. Bell's theorem is exactly the statement that no seed reproduces the correlations across *all* setting pairs. If you could write a deterministic function f(λ, setting) → ±1 matching quantum predictions, you would violate a counting bound. This is the single sharpest place where a systems intuition must be surrendered rather than refined.

Key takeaways

- Superposition is basis-relative: every state is a basis state in some basis. Entanglement is basis-independent and cannot be transformed away.
- Amplitudes cancel; probabilities do not. That difference is the source of quantum advantage.
- Entangled subsystems carry no individual information; the information is in the correlation. No-signaling holds — a classical channel is still required.
- EPR is a valid argument from locality plus realism to incompleteness. Bell showed those premises are experimentally refutable.
- CHSH: local realism gives $S \leq 2$; quantum mechanics gives $2\sqrt{2}$. Experiment confirms $2\sqrt{2}$, loophole-free since 2015.

Self-check

Q1 *(Objective 2)* — A colleague proposes entanglement-based faster-than-light messaging: Alice's basis choice instantly affects Bob's state. Identify the precise step that fails, and state what Bob would have to observe for the scheme to work.

Q2 *(Objective 2)* — Bell's inequality derivation makes no reference to quantum mechanics. Explain why that is the source of its force, and name the analogous property of a proof about a distributed protocol that makes it worth more than a benchmark.

A1 · A2 — Check your answers

A1 — The failure is at "affects Bob's state," where "state" equivocates between the joint description and Bob's **reduced density matrix** — the only thing Bob can access.

Before Alice acts, Bob's marginal is $\rho_B = I/2$: maximally mixed, 50/50 in every basis. After Alice measures in *any* basis, Bob's marginal is still exactly $I/2$. Alice's choice changes the joint description and changes what Alice can *predict* about Bob's outcomes, but it does not change any distribution Bob can sample. Bob sees uniform noise either way, and no measurement or statistical test on his qubits alone distinguishes the cases.

For the scheme to work, Bob would have to observe a **change in his marginal distribution** conditioned on Alice's choice. The no-signaling theorem proves that cannot happen for any entangled state under any local operation — it follows from the partial trace commuting with operations on the other subsystem. The correlation is real, but it is only visible in the *joint* record, and assembling that record requires Alice to send her results over a classical channel at $\leq c$.

The compact version: entanglement is correlation without communication. Correlation you cannot see until you compare notes carries no signal.

A2 — Its force comes from being an **impossibility result about a class of models**, not a comparison between two specific theories. Bell assumes only that outcomes are determined by some variables λ carried by the particles, and that a distant setting choice does not affect local outcomes. He derives $S \leq 2$ from those assumptions alone. Any theory in that class — existing, proposed, or never conceived — is bound by it.

This is why the experimental violation is decisive rather than merely favorable to quantum mechanics. It does not show quantum mechanics beat a competitor; it shows an entire *category* of explanation is unavailable. Even if quantum mechanics is superseded, the successor cannot be local-realist.

The distributed-systems parallel is a proof like FLP or the CAP theorem: a bound derived from the model's assumptions rather than measured from an implementation. A benchmark tells you what one system did on one workload, and a faster machine invalidates it. FLP tells you no deterministic protocol achieves consensus with one faulty process in an asynchronous system — and no amount of engineering repeals it. Bell is that kind of result, which is why it settled a thirty-year argument that no experiment comparing predictions could have settled.